

Chapter 6

Central banking

Learning objectives

- To understand the role of central banks in the financial sector
- To describe the main functions of a central bank
- To understand the monetary policy functions of central banks
- To discuss the arguments for and against an independent central bank
- To understand the relevance of central banks during crises

6.1 Introduction

The core functions of central banks in any countries are to manage monetary policy with the aim of achieving **price stability**; to prevent liquidity crises, situations of money market disorders and financial crises; and to ensure the smooth functioning of the payments system. This chapter explores these issues and focuses in particular on the conduct of monetary policy, distinguishing between instruments, targets and goals. Furthermore, it examines some basic concepts as they relate to central banking theory. Specifically, the chapter investigates the following fundamental areas:

- What are the monetary policy functions of a central bank?
- Why do banks need a central bank?
- Should central banks be independent from government?

It is important to note that, in addition to monetary policy objectives, the majority of central banks operate under the presumption that they have a policy responsibility for **financial stability**. For example, the Bank of England's primary functions are to maintain monetary stability and oversee financial stability of the UK financial system. In addition, the Bank of England acts as the lender of last resort and as the custodian of the official gold reserves in the United Kingdom. The focus of this chapter is on the monetary policy function of central banks. Issues relating to financial stability will be discussed in Chapter 8.

6.2 What are the main functions of a central bank?

A central bank can generally be defined as a financial institution responsible for overseeing the monetary system for a nation, or a group of nations, with the goal of fostering economic growth without inflation. The main functions of a central bank can be listed as follows:

- 1 The central bank controls the issue of notes and coins (legal tender). Usually, the central bank will have a monopoly of the issue, although this is not essential as long as the central bank has power to restrict the amount of private issues of notes and coins.
- 2 It has the power to control the amount of credit-money created by banks. In other words, it has the power to control, by either direct or indirect means, the money supply.
- 3 A central bank should also have some control over non-bank financial intermediaries that provide credit.
- 4 Encompassing both points 2 and 3, the central bank should effectively use the relevant tools and instruments of monetary policy in order to control:
 - credit expansion;
 - liquidity; and
 - the money supply of an economy.
- 5 The central bank should oversee the financial sector in order to prevent crises and act as a **lender of last resort (LOLR)** in order to protect depositors, prevent widespread panic withdrawals and otherwise prevent the damage to the economy caused by the collapse of financial institutions.
- 6 A central bank acts as the government's banker. It holds the government's bank account and performs certain traditional banking operations for the government, such as deposits and lending. In its capacity as banker to the government it can manage and administer the country's national debt.
- 7 The central bank also acts as the official agent to the government in dealing with all its gold and foreign exchange matters. The government's reserves of gold and foreign exchange are held at the central bank. A central bank, at times, intervenes in the foreign exchange markets at the behest of the government in order to influence the exchange value of the domestic currency.

The first central banks were created in the 17th century (the Amsterdamse Wisselbank, the Swedish Riksbank and the Bank of England). However, the dawn of central banks was in the 19th century, as shown in Figure 6.1.

Central banks differ from country to country in their organisation and structure as well as in the specific tasks carried out; their main tasks nowadays is to carry out a country's monetary policy function. Central banks usually also have important financial stability functions, which become more prominent during times of financial turmoil. As central banks' mandates in the economy expanded in the aftermath of the financial crisis, new challenges arose in particular with respect to the balance between their **accountability** and *independence* as well as the increased risk of politicisation. Section 6.3 briefly introduces monetary policy objectives.

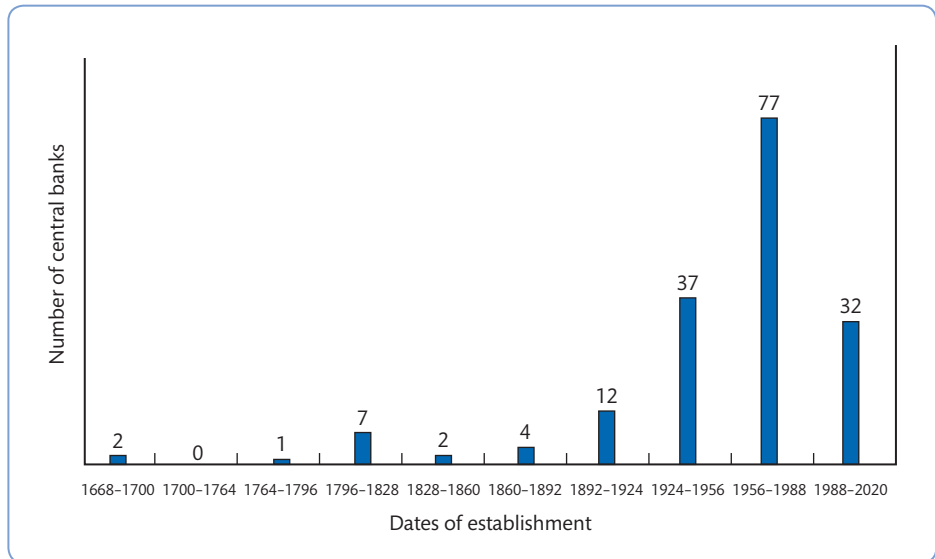


Figure 6.1 Dates of establishment of central banks

Source: Adapted from Bossu and Rossi (2019).

6.3 How does monetary policy work?

There are five major forms of economic policy (or, more strictly macroeconomic policy) conducted by governments that are of relevance. These are monetary policy, fiscal policy, exchange rate policy, prices and incomes policy, and national debt management policy.

- **Monetary policy** is concerned with the actions taken by central banks to influence the availability and cost of money and credit by controlling some measure (or measures) of the money supply and/or the level and structure of interest rates.¹
- **Fiscal policy** relates to changes in the level and structure of government spending and taxation designed to influence the economy. As all government expenditure must be financed, these decisions also, by definition, determine the extent of public sector borrowing or debt repayment. An expansionary fiscal policy means higher government spending relative to taxation. The effect of these policies would be to encourage more spending and boost the economy. Conversely, a contractionary fiscal policy means raising taxes and cutting spending.
- **Exchange rate policy** involves the targeting of a particular value of a country's currency exchange rate thereby influencing the flows within the balance of payments. In some countries it may be used in conjunction with other measures such as exchange controls, import tariffs and quotas.²

¹ See also Appendix A1.

² Note that exchange controls, tariffs and quotas are restricted or forbidden under a number of trade agreements (such as those implemented by the EU and the World Trade Organization).

- A **prices and incomes policy** is intended to influence the inflation rate by means of either statutory or voluntary restrictions upon increases in wages, dividends and/or prices.
- **National debt management policy** is concerned with the manipulation of the outstanding stock of government debt instruments held by the domestic private sector with the objective of influencing the level and structure of interest rates and/or the availability of reserve assets to the banking system.

In this section we focus on what monetary policy involves. However, it must be remembered that any one policy mentioned above will normally form part of a policy package, and that the way in which that policy is employed will be dependent upon the other components of that package. Box 6.1 provides essential background reading for this section on the concept and functions of **money**.

BOX 6.1 THE CONCEPT AND FUNCTIONS OF MONEY AND MONETARY AGGREGATES

In general, money is represented by the coins and notes that we use in our daily lives; it is the commodity readily acceptable by all people wishing to undertake transactions. It is also a means of expressing a value for any kind of goods or service. For economists, money is referred to as 'money supply' and includes anything that is accepted in payment for goods and services or in the repayment of debts. In an economic system money serves four main functions: (1) medium of exchange; (2) unit of account; (3) store of value; and (4) standard of deferred payment.

- 1 *Medium of exchange* is probably the main function of money. If barter were the only type of trade possible, there would be many situations in which people would not be able to obtain the goods and services that they wanted most. The advantage of the use of money is that it provides the owner with generalised purchasing power. The use of money gives the owner flexibility over the type and quantities of goods they buy, the time and place of their purchases, and the parties with whom they choose to deal. A critical characteristic of a medium of exchange is that it be acceptable as such. It must be readily exchangeable for other things. It is usual for the government to designate certain coins or paper currency as the medium of exchange.
- 2 If money is acceptable as a medium of exchange it almost certainly comes to act as a *unit of account* by which the prices of all commodities can be defined and then compared. This, of

course, simplifies the task of deciding how we wish to divide our income between widely disparate items. For this reason, it is sometimes said that money acts as a measure of value, and this is true if value is taken to mean both price and worth, the latter being a much more subjective concept.

- 3 Money is also a liquid *store of value* in that it provides individuals with a means of holding and accumulating their wealth in a form which can, at any time, be converted immediately into goods and services. When a person holds money as a store of value they are effectively treating it as a substitute to holding alternative forms of financial assets such as bonds or deposit accounts. The holder of money therefore foregoes the payment of an explicit yield.
- 4 Money can also act as a *standard of deferred payment*. Due to this function it is possible to undertake a number of transactions in the present and actually settle the account (or bill) at some time in the future, e.g. buy now and pay later. The sale and production of goods are made easier by money performing this function since goods and services can be acquired prior to payment being made. Because money acts as a standard of deferred payment the acquisition of labour, raw materials and other goods and services can be acquired, and the various parties involved will know the sums involved and payments to be made at a future date. Although this



Box 6.1 The concept and functions of money and monetary aggregates (continued)

particular function of money is not essential for lending, borrowing and production to take place, it certainly makes such activities easier. Money's function as a standard for deferred payment may be questioned in times of high inflation where the real value of money declines rapidly. In such situations the debtor would benefit from a deferred payment. However, the meaning of the function of money as a standard for deferred payment is that it permits commercial lending to take place. A borrower can agree that if a lender supplies them with ten units of money today, they will pay back eleven units in (say) three months' time. The charging of interest has become possible.

The formal definition of what constitutes money is summarised by central banks' definitions of what they call **monetary aggregates**.

Bank of England

In the UK, for example, official estimates of the money supply have been published since 1966. The earliest definition of the money supply was a broad one covering notes and coin held by UK non-banks and deposits (in both sterling and foreign currency) held by UK residents with banks in the UK. From 1970 onwards, this definition has been amended and supplemented on a number of occasions, reflecting developments in the financial system and policy. Common practice among central banks is to construct monetary aggregates from a list of monetary assets by adding together those that are considered to be likely sources of monetary services. The monetary aggregates play a role in the formulation of the monetary policy. In the UK, M0 is the narrowest measure of money stock and M4 the broadest.

M0 = sterling notes and coins

+ banks' operational deposits with the Bank of England

M4 = UK private sector other than MFIs holdings of:

+ sterling notes and coins
+ sterling deposits, including certificates of deposit
+ commercial paper, bonds, floating rate notes (FRNs) and other instruments up to and including five years' original maturity issued by UK MFIs

+ claims on UK MFIs arising from repos
+ estimated holdings of sterling bank bills
+ 35 per cent of the sterling inter-MFI difference

Note that MFIs comprise the central bank (i.e. banking department and issue department of the Bank of England), other banks and building societies (see also Section 2.3). The inter-MFI difference is the difference between MFIs' liabilities attributed to other MFIs and MFIs' assets attributed to other MFIs. In theory, these items should be equal but in practice small reporting errors cause a difference. Deposits include sight deposits that are funds that can be converted immediately and without restrictions into cash; and time deposits, that are funds invested at a bank for a fixed period of time and that the investor cannot access until the end of an agreed period (e.g. 30- and 60-day savings accounts). Certificates of deposit (CD) are negotiable certificates confirming that a (usually large) deposit has been made for a specified period of time with a bank.

M4 can be analysed either in terms of its components – cash and deposits – or of its asset counterparts, which represent the other side of the banks' and building societies' balance sheets (and must, as an accounting identity, sum to the same). These counterparts include banks' and building societies' lending to the private sector and their transactions with the public sector and with overseas residents.

European Central Bank

The European Central Bank's definition of monetary aggregates is slightly different from the Bank of England's detailed above. In line with international practice, the euro-system has defined a narrow aggregate (M1), an 'intermediate' aggregate (M2) and a broad aggregate (M3). These aggregates differ with regard to the degree of 'moneyness' of the assets included. These are defined as:

- *Narrow money* (M1) includes currency (i.e. bank notes and coins), as well as balances which can immediately be converted into currency or used for cashless payments (i.e. overnight deposits).
- *Intermediate money* (M2) comprises narrow money (M1) and deposits with a maturity of up to two years and deposits redeemable at a period



Box 6.1 The concept and functions of money and monetary aggregates (*continued*)

of notice of up to three months. The definition of M2 reflects the particular interest in analysing and monitoring a monetary aggregate that, in addition to currency, consists of deposits which are liquid.

- *Broad money* (M3) comprises M2 and marketable instruments issued by the MFI sector. Certain money market instruments, in particular money

market fund (MMF) shares/units and repurchase agreements are included in this aggregate. A high degree of liquidity and price certainty make these instruments close substitutes for deposits. As a result of their inclusion, M3 is less affected by substitution between various liquid asset categories than narrower definitions of money and is therefore more stable.

Monetary policy relates to the control of some measure (or measures) of the money supply and/or the level and structure of interest rates. Much greater emphasis nowadays is placed on monetary policy within a government's policy package. This is because a broad consensus has emerged that suggests that price stability is an essential pre-condition for achieving the central economic objective of high and stable levels of growth and employment. Monetary policy is viewed as the preferred policy choice for influencing prices.

Although traditionally the choice of monetary policy over fiscal policy as the main policy tool was viewed as a matter of ideological choice, nowadays it is seen more as a pragmatic solution. As it is widely recognised that high and variable inflation harms long-term growth and employment, policy makers have tended to focus on those policies that appear to be most successful in dampening inflationary pressures. Price stability, therefore, has become a key element of economic strategy, and monetary policy is widely accepted as the most appropriate type of policy to influence prices and price expectations.

The preference for using monetary policy over other types of policy relates to two main factors – the role of the monetary authorities (central banks) as sole issuers of banknotes and bank reserves (known as the monetary base) and the long-run neutrality of money (see below).

The central bank is the monopoly supplier of the monetary base and as a consequence can determine the conditions at which banks borrow from the central bank. The central bank can influence liquidity in the short-term money markets and so can determine the conditions at which banks buy and sell short-term wholesale funds. By influencing short-term money market rates, the central bank influences the price of liquidity in the financial system and this ultimately can impact on various economic variables such as output or prices.

In the long run a change in the quantity of money in the economy will be reflected in a change in the general level of prices but it will have no permanent influence on real variables such as the level of (real) output or unemployment. This is known as the *long-run neutrality of money*. The argument goes that real income, or the level of employment is, in the long term, determined solely by real factors, such as technology, population growth or the preferences of economic agents. Inflation is therefore solely a monetary phenomenon.

As a consequence, in the long run:

- a central bank can only contribute to raising the growth potential of the economy by maintaining an environment of stable prices;
- economic growth cannot be increased through monetary expansion (increased money supply) or by keeping short-term interest rates at levels inconsistent with price stability.

In the past it has been noted that long periods of high inflation are usually related to high monetary growth. While various other factors (such as variations in aggregate demand, technological changes or commodity price shocks) can influence price developments over the short period, over time these influences can be offset by a change in monetary policy.

6.4 Monetary policy functions of a central bank

The most important function of any central bank is to undertake monetary control operations. Typically, these operations aim to administer the amount of money (money supply) in the economy and differ according to the monetary policy objectives they intend to achieve. These latter are determined by the government's overall macroeconomic policies (see Box 6.2).

BOX 6.2 MONETARY POLICY OBJECTIVES

Monetary policy is one of the main policy tools used to influence interest rates, inflation and credit availability through changes in the supply of money (or liquidity) available in the economy. It is important to recognise that monetary policy constitutes only one element of an economic policy package and can be combined with a variety of other types of policy (e.g. fiscal policy) to achieve stated economic objectives. Historically, monetary policy has, to a certain extent, been subservient to fiscal and other policies involved in managing the macro economy, but nowadays it can be regarded as the main policy tool used to achieve various stated economic policy objectives (or goals). These vary across central banks. The main objectives of economic (and monetary) policy include:

- *High employment* – often cited as a major goal of economic policy. Having a high level of unemployment results in the economy having idle resources that result in lower levels of production and income, lower growth and possible social unrest. However, this does not necessarily mean that zero unemployment is a preferred policy goal. A certain level of unemployment is often felt to be necessary for the efficient operation of a dynamic economy. It will take people a period of time to switch between jobs, or retrain for new jobs, and so on – so even near full employment there may be people switching jobs who are temporarily out of work. This is known as *frictional unemployment*. In addition, unemployment may be a consequence of mismatch in skills between workers and what employers want – known as *structural unemployment*. (Typically, although structural unemployment is undesirable monetary policy cannot alleviate this type of unemployment). The goal of high employment, therefore, does not aim to achieve zero unemployment but seeks to obtain a level above zero that is consistent with matching the demand and supply of labour. This level is known as the *natural rate of unemployment*. Milton Friedman (1968) defined the natural rate of unemployment as the level of unemployment which resulted from real economic forces, the long run level of which could not be altered by monetary policy. There is a large economic literature on measuring the natural rate of unemployment, although with little consensus on a measure, as shifts in local labour market conditions, changes in both labour supply and labour demand can influence the natural rate of unemployment.
- *Price stability* – considered an essential objective of economic policy, given the general wish to avoid the costs associated with inflation. Price stability is viewed as desirable because rising price levels create uncertainty in the economy, and this can adversely affect economic growth. Many economists (but by no means all) argue that low inflation is a necessary prerequisite for achieving sustainable economic growth. Some central banks (e.g. the European Central Bank) list price stability as a primary objective. A formal adoption of an inflation target is one way to control the general rise in the



Box 6.2 Monetary policy objectives (*continued*)

price level. For more information on how inflation targeting works, see e.g. Hammond (2012).

- *Stable economic growth* – provides for the increases over time in the living standards of the population. The goal of steady economic growth is closely related to that of high employment because firms are more likely to invest when unemployment is low – when unemployment is high, and firms have idle production they are unlikely to want to invest in building more plants and factories. The rate of economic growth should be at least comparable to the rates experienced by similar nations.
- *Financial market stability* – also an important objective of public policy and one major concern of central banks. A collapse of financial markets can have major adverse effects on an economy. The crisis that hit the global financial system in 2007–2009 caused a sharp contraction in output, weakened the effectiveness of the transmission mechanism and contributed to inefficient capital allocation. A similar crisis occurred in 1929, when the US Wall Street Crash resulted in a fall of manufacturing output by 50 per cent and an increase in unemployment to 25 to 30 per cent of the US workforce by 1932. (Over 11,000 banks closed over this period.) Such major crises may be rare, but they do highlight the serious consequences of financial crises. A less dramatic example of how policy makers view the adverse effects of a crisis in financial markets relates to the case of the US-based hedge fund Long-Term Capital Management. In September 1998 the US Federal Reserve organised a rescue of Long-Term Capital Management, a very large and prominent hedge fund on the brink of failure. The monetary authorities intervened because they were concerned about the serious consequences for world financial markets if they allowed the hedge fund to fail. Note that financial market stability is influenced by stability of interest rates because increases in interest rates can lead to a decrease in the value of bonds and other investments resulting in losses in the holders of such securities. In early 2020, the fast spread of Covid-19 delivered an unprecedented macro-economic shock, bringing international uncertainty and pushing the world economy into a recession much faster than any previous crisis. The shutdown of the bulk of economic activity has placed the financial system under strain, so prompt actions have been taken by the authorities to monitor financial stability risks.
- *Stability in foreign exchange markets* – has become a policy goal of increasing importance especially in the light of greater international trade in goods, services and capital. A rise in the value of a currency makes exports more expensive (an increase in the value of sterling relative to the dollar means that consumers in the US have to pay more for UK goods), whereas a decline in the value of a currency leads to domestic inflation (if sterling declines relative to the dollar, US goods sold in the UK become more expensive). Extreme adverse movements in a currency can therefore have a severe impact on exporting industries and can also have serious inflationary consequences if the economy is open and relatively dependent on imported goods. Ensuring the stability of foreign exchange markets is therefore seen as an appropriate goal of economic policy.

At first glance it may appear that all these policy objectives are consistent with one another, but conflicts do arise. The objective of price stability can conflict with the objectives of interest rate stability and full employment (at least in the short run) because as an economy grows and unemployment declines, this may result in inflationary pressures forcing up interest rates. If the monetary authorities do not let interest rates increase this could fuel inflationary pressures, yet if they do increase rates then unemployment may occur. These sorts of conflicts create difficulties for the authorities in conducting monetary and other macroeconomic policy.

Typically, the most important long-term monetary target of a central bank is price stability that implies low and stable inflation levels. As shown in Figure 6.2 such a long-term

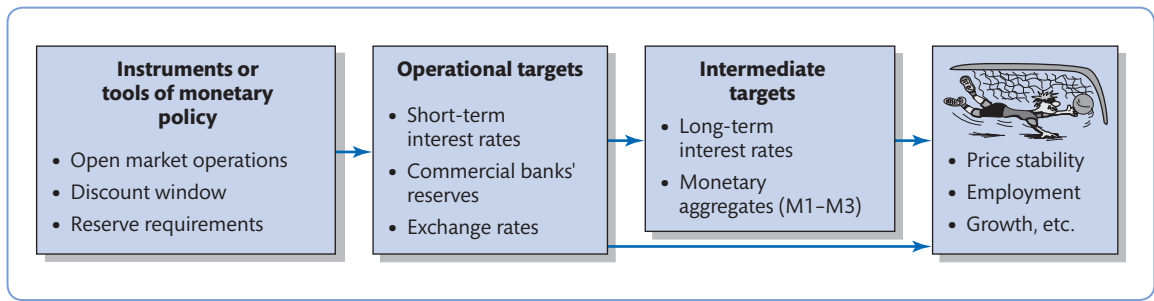


Figure 6.2 Monetary policy instruments, targets and goals

goal can only be attained by setting short-term operating targets. Operating targets are usually necessary to achieve a particular level of interest rates, commercial banks' reserves or exchange rates. Often, they are complemented by intermediate targets such as a certain level of long-term interest rates or broad money growth (monetary aggregates). In choosing the intermediate targets, policy makers should take into account the stability of money demand and the controllability of the monetary aggregate. The chosen target should also be a good indicator of the effect of the monetary policy decision on the price stability target. Broad aggregates normally show higher stability and display better indicator properties than narrow aggregates. In contrast, in the short term narrow aggregates are easier to control via official interest rates than broad aggregates. Although central banks cannot use monetary policy instruments directly to affect intermediate targets, they can use them to affect operating targets, such as reserve money and short-term interest rates, which influence movements in intermediate variables.

Let us now focus on the tools or instruments of monetary policy. In the past, it was common for central banks to exercise direct controls on bank operations by setting limits either to the quantity of deposits and credits (e.g. ceilings on the growth of bank deposits and loans), or to their prices (by setting maximum bank lending or deposit rates). As a result of the significant financial liberalisation process aimed at achieving an efficient allocation of financial resources in the economy, there has been a movement away from direct monetary controls towards indirect ones (Gray and Talbot, 2006).

Indirect instruments influence the behaviour of financial institutions by affecting initially the central banks' own balance sheet. In particular the central bank will control the price or volume of the supply of its own liabilities (reserve money) that in turn may affect interest rates more widely and the quantity of money and credit in the whole banking system.

To understand how a central bank can direct and control the money supply through indirect instruments it is essential to first evaluate the role and importance that the different classes of assets and liabilities detailed in its balance sheet have for monetary policy.

As any other bank, the central bank has to produce a financial statement each year (for more details on banks' accounts see Chapter 9) and the items contained in its balance sheet do not differ substantially from those of commercial banks. Table 6.1 illustrates a simplified central bank balance sheet.

The asset side of the balance sheet of any central bank includes two asset types: net foreign reserves and domestic claims. These are typically represented by loans and advances (to banks and other institutions) and debt securities. On the liability side, it is possible to identify the so-called 'high powered' money or monetary base (currency and reserves), other deposits

Table 6.1 A simplified central bank balance sheet

Assets	Liabilities
Foreign assets	Reserve money
Domestic assets	Currency in circulation
Claims on government and public enterprises	Reserves of commercial banks
Claims on the private sector	Foreign liabilities
Claims on domestic money banks	Other deposits of commercial banks
Claims on other financial sector entities	Central bank securities
	Government deposits
	Others
	Equity capital
Total	Total

Source: Adapted from Filardo and Yetman (2011), Table 1.

of banks and other institutions (e.g. the government), the central bank's own securities and equity capital. This latter includes accumulated profits/losses and transfers of resources from the government.

Central banks' balance sheets have expanded markedly. Financial crises and economic downturns have resulted in unusual and/or extraordinary measures that are reflected in substantial changes in both the structure and size of central banks' balance sheets. In the UK, the money market reforms of 18 May 2006 and the measures implemented after the 2007–2009 crisis had a substantial effect on the Bank of England's balance sheet. 'Unprecedented' expansion trends in central banks' balance sheets are found both in the US and eurozone and in Asia Pacific, with China leading the trend. In most cases changes have been driven by the growth in net foreign assets, particularly US\$ denominated bonds. Table 6.2 reports selected balance sheet data for the Bank of England (as published on 29 February 2019).

If we focus on the asset side, it is possible to see that the item 'other loans and advances' represents the most significant asset of the Bank of England (73.97 per cent) while on the liability side, deposits from banks and other financial institutions are the largest proportion (82.95 per cent). These latter are the reserve balances, i.e. current account balances held by commercial banks at the Bank of England.

As discussed in Box 6.3, currently commercial banks earn an interest on their reserves that is equal to the Bank official rate and this represents a key part in the implementation of the UK's monetary policy (Hauser, 2019). Since March 2009 there has been a sharp increase in reserves balances and this reflects the fact that asset purchases under the Monetary Policy Committee (MPC)'s policy of **quantitative easing (QE)** have been financed by increasing reserves balances (see Section 6.4.5 for a specific discussion of unconventional monetary policy tools). To steer the quantity towards the required target, the Bank of England uses its open market operations.

The indirect instruments used by central banks in monetary operations are generally classified into:

- **open market operations** (OMOs);
- **discount window** (also known as **standing facilities**); and
- **reserve requirements**.

Table 6.2 Balance sheet, Bank of England

	(£m)
Assets	
Cash and balances with other central banks	698
Loans and advances to banks and other financial institutions	136,222
Other loans and advances	445,002
Securities held at fair value through profit and loss	6,064
Derivative financial instruments	156
Available for sale securities	–
Securities held at amortised cost	–
Securities held at fair value through other comprehensive income	10,302
Inventories	5
Property, plant and equipment	412
Intangible assets	47
Retirement benefit assets	927
Other assets	499
Total assets (100%)	601,583
Liabilities	
Deposits from central banks	9,922
Deposits from banks and other financial institutions	495,406
Deposits from banks – cash ratios deposits	7,884
Other deposits	76,799
Foreign currency bonds in issue	6,042
Derivative financial instruments	111
Current tax liabilities	–
Deferred tax liabilities	289
Retirement benefit liabilities	207
Other liabilities	573
Total liabilities (100%)	597,233
Equity	
Capital	15
Retained earnings	3,076
Other reserves	1,259
Total equity attributable to shareholder	4,350
Total liabilities and equity	601,583

Source: Bank of England (2019) *Annual Report and Accounts*.

BOX 6.3 BANK OF ENGLAND AND THE OFFICIAL INTEREST RATE

The MPC has responsibility within the Bank of England for formulating monetary policy. The MPC announces the interest rate decision the first Thursday of every month at 12 noon, reviewing the short-term **official bank rate** (also referred to as 'policy rate' or 'base rate') in response to economic conditions. In the United Kingdom, the bank rate (0.1 per cent since 19 March 2020) is the overnight interest rate at which the Bank of England lends to financial institutions. In setting this key rate, the Bank considers the amount at which banks borrow from each other overnight.

The official bank rate replaces the repo rate that was used prior to the May 2006 reforms to the Bank's framework for its operations in the sterling money market. These changes allow banks and building societies that hold cash ratio deposits (CRDs) at the Bank of England to place additional deposits overnight in an account at the central bank.^a These assets are called reserve balances and they represent claims on the central bank. Their role includes the following: (1) they are the most liquid risk-free assets in the economic system, together with banknotes; (2) they are the ultimate assets for settling payments between customers of different banks; and (3) they facilitate banks' liquidity management that derives from typical lending and borrowing bank activities. In recent years, as a result of the quantitative easing, the holding of reserves has increased significantly. The Bank invests the CRDs in interest-yielding assets, generating income to fund its policy functions. Any gains or losses will be recognised in the Bank's capital.

Since 2009 the Bank of England pays reserves balances an interest that is typically the bank rate. By doing so the Bank keeps market interest rates in line with the bank rate and establishes an important benchmark short-term risk-free rate. Banks can decide to change their holdings of reserves daily to meet day to day liquidity needs. The bank rate will affect the rates banks are willing to charge or pay on short-term loans or borrowing in the money market. In the longer term, money markets rates will be influenced by expectations about future bank rates. Consumer and business demand (i.e. the aggregate spending) will be affected by changes in the bank rate through several channels in the transmission

mechanism of monetary policy, namely by changes in (1) the deposit and lending rates charged by commercial banks and building societies, (2) financial assets prices (e.g. bonds and equities) and (3) exchange rates (for more details, see Box 6.5).

In the context of the monetary framework operations of the Bank of England, operational standing facilities (OSFs) have two roles: (1) to provide an arbitrage mechanism in normal market conditions to prevent money market rates moving far away from the Bank rate (and therefore they are a vital part of implementing monetary policy) and (2) to provide a means for participating banks to manage unexpected payment shocks which may arise due to technical problems in banks' own systems or in the market-wide payments and settlements infrastructure.

The OSFs allow participating institutions to deposit reserves with or borrow reserves directly from the Bank on a bilateral basis throughout each business day. The operational standing lending facility takes the form of an overnight repo transaction against high-quality, highly liquid collateral. The operational standing deposit facility takes the form of an unsecured deposit with the Bank. On those terms, the OSFs are available in unlimited size. Commercial banks borrowing from the lending facility are required to pay a premium over bank rate, while those placing reserves in the deposit facility are remunerated at a rate below bank rate. Commercial banks will typically be unwilling to deal in the market on worse terms than those available at the Bank – for example, if market rates are above the lending facility rate, banks will tend to borrow from the Bank in preference to the market. So the OSF rates establish a corridor around bank rate and help limit volatility in overnight market interest rates while incentivising banks to manage their liquidity prudently.

Note: ^a Cash ratio deposits (CRDs) are non-interest bearing deposits lodged with the Bank of England by eligible institutions (i.e. banks and building societies), who have reported average eligible liabilities (ELs) of over £600 million over a calculation period. The level of each institution's CRD is currently calculated twice-yearly (currently in May and November) at 0.30 per cent (up from 0.18 per cent in the previous period) of average ELs, over the previous six end-calendar months, in excess of £600mn. The value bands and ratios were specified by HM Treasury in the Cash Ratio Deposits (Value Bands and Ratios) Order 2018 No 633.

Open market operations are by far the most common means of implementing conventional monetary policy and they are used by central banks all over the world. By injecting (or absorbing) liquidity in (or from) the banking sector central banks manipulate the short-term interest rate and the money supply in the economy. However, there are variations in the way central banks use these instruments and conduct these operations across countries. For useful summary tables comparing frameworks across countries see Bank for International Settlements (2019b).

6.4.1 Debt securities and open market operations

Debt securities are mainly represented by Treasury securities (i.e. government debt) that central banks use in open market operations. These operations are the most important tools by which central banks can influence the amount of money in the economy.

Although the practical features of open market operations may vary from country to country, the principles are the same: the central bank operates in the market and purchases or sells government debt to the non-bank private sector. In general, if the central bank sells government debt the money supply falls (all other things being equal) because money is taken out of bank accounts and other sources to purchase government securities. This leads to an increase in short-term interest rates. If the government purchases (buys-back) government debt this results in an injection of money into the system and short-term interest rates fall. As a result, the central bank can influence the portfolio of assets held by the private sector. This will influence the level of liquidity within the financial system and will also affect the level and structure of interest rates (see Figure 6.3).

The main attractions of using open market operations to influence short-term interest rates are as follows:

- they are initiated by the monetary authorities who have complete control over the volume of transactions;
- open market operations are flexible and precise – they can be used for major or minor changes to the amount of liquidity in the system;
- they can easily be reversed;
- open market operations can be undertaken quickly.

Open market operations are the most commonly used indirect instruments of monetary policy in developed economies. One of the main reasons for the widespread use of open

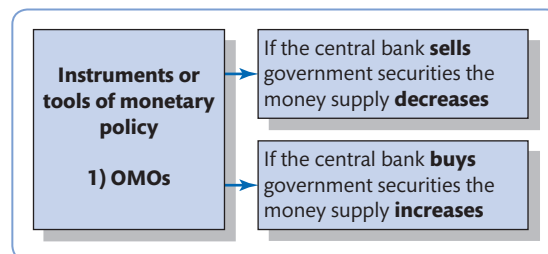


Figure 6.3 Open market operations: outright purchases of government securities

market operations relates to their flexibility in terms of both the frequency of use and scale (i.e. quantity) of activity. These factors are viewed as essential if the central bank wishes to fine-tune its monetary policy. In addition, OMOs have the advantage of not imposing a tax on the banking system.

In the United Kingdom, the Bank of England uses open-market operations (outright buying and selling of securities) to supply, in aggregate, the reserves that banks need to meet their collected targets. However, the amount of reserves that it aims to supply is affected by the banks' reserves targets as well as the expected impact of other factors that may influence the supply, such as an increase in the demand for banknotes (Bank of England, 2012b). The Bank of England may also supply banks with funds temporarily by using 'repo' agreements, i.e. sale and repurchase agreements relating to financial assets. For example, it may engage in gilt repo transactions, whereby the Bank purchases gilt-edged securities from private sector counterparties with a legally binding commitment that the securities will be repurchased by the counterparties at a pre-determined price and date. Gilt repos are, in effect, cash loans with the gilt-edged securities used as collateral. While the stance of monetary policy is expressed as the level of the Bank Rate (see Box 6.3), under the current operational framework two types of open market operations can be used to supply reserves: short-term and long-term. The former are aimed to ensure that the Bank of England rate does not diverge from the interbank rate and to steer the quantity of reserves to the amount necessary for the banking system as a whole to meet its targets. The latter have been introduced to provide a sort of liquidity insurance as the Bank of England offers to lend reserves for longer periods against a broad range of collateral (Clews *et al.*, 2010).

6.4.2 Loans to banks and the discount window

The second most important monetary policy tool of a central bank is the so-called discount window (in the United Kingdom this tool is often referred to as 'standing facilities'). It is an instrument that allows eligible banking institutions to borrow money from the central bank, usually to meet short-term liquidity needs. By changing the discount rate, that is, the interest rate that monetary authorities are prepared to lend to the banking system, the central bank can control the supply of money in the system. If, for example, the central bank is increasing the discount rate, it will be more expensive for banks to borrow from the central bank so they will borrow less thereby causing the money supply to decline. Vice versa, if the central bank is decreasing the discount rate, it will be cheaper for banks to borrow from it so they will borrow more money (Figure 6.4).

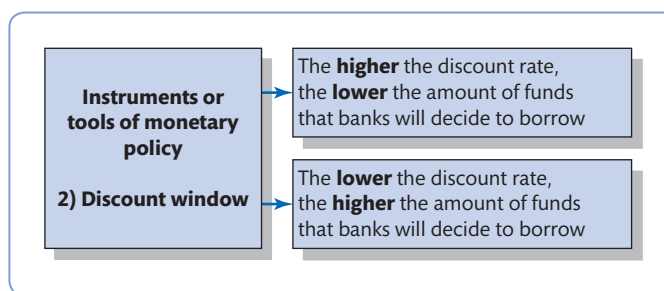


Figure 6.4 Discount window

Manipulation of the discount rate can therefore influence short-term rates in the market. For instance, the eurozone's discount rate is known as a 'marginal lending facility', which offers overnight credit to banks from the euro system. Table 6.3 summarises the Eurosystem monetary policy operations.

In the United States, when the Federal Reserve System was established in 1913, lending reserve funds through the discount window was intended to be the most important instrument of central banking operations but it was soon replaced by open market operations. Indeed, today banks are discouraged from using this type of borrowing.³ Direct lending to banks can also occur through the central bank's lender of last resort (LOLR) function. By acting as a lender of last resort the central bank provides liquidity support directly to individual financial institutions if they cannot obtain finance from other sources. Therefore, it can help to prevent financial panics. Some drawbacks of this function are discussed in Section 6.5.1.

Table 6.3 Eurosystem monetary policy operation

Monetary policy operations	Types of instruments		Maturity	Frequency	Procedure	
	Liquidity providing	Liquidity absorbing				
Open market operations						
Main refinancing operations	● Reverse transactions		1 week	Weekly	Standard Tenders	
Longer-term refinancing operations	● Reverse transactions		3 months	Monthly	Standard Tenders	
Fine tuning operations	● Reverse transactions	● Reverse transactions	Non-standardised	Non-regular	Quick tenders	
	● Foreign exchange swaps	● Collection of fixed-term deposits				Bilateral procedures
		● Foreign exchange swaps				
Structural operations	● Reverse transactions	● Issuance of debt certificates	Standardised/ non-standardised	Regular and non-regular	Standard tenders	
	● Outright purchases	● Outright sales		Non-regular	Bilateral procedures	
Standing facilities						
Marginal lending facility	Reverse transactions		Overnight	Access at the discretion of counterparties		
Deposit facility	Deposits		Overnight	Access at the discretion of counterparties		
Minimum reserves*						
The Eurosystem requires credit institutions to hold minimum reserves on accounts with the national central banks. Until January 2012 the reserve ratio was a minimum of 2 per cent for the majority of the items for which the reserve base applies (deposits, debt securities and money market paper). Since then, the ratio has been lowered to 1 per cent. Reserve holdings are remunerated at the Eurosystem's rate on its main refinancing operations.						

Source: www.ecb.europa.eu/mopo/implementation/html/index.en.html See also The Governing Council of the European Central Bank (2018). Published on the Official Journal of the European Union (L 95/123 of 13/04/2018).

Note: *Regulation (EC) No. 1745/2003 of the European Central Bank of 12 September 2003 on the application of minimum reserves (ECB/2003/9) (OJ L 250, 2.10.2003, pp. 10–16).

³For more details, see www.frbdiscountwindow.org

6.4.3 Reserve requirements

Banks need to hold a quantity of reserve assets for prudential purposes. If a bank falls to its minimum desired level of reserve assets it will have to turn away requests for loans or else seek to acquire additional reserve assets from which to expand its lending. The result in either case will generally be a rise in interest rates that will serve to reduce the demand for loans.

The purpose of any officially imposed reserve requirements is effectively to duplicate this process. If the authorities impose a reserve requirement in excess of the institutions' own desired level of reserves (or else reduce the availability of reserve assets) the consequence will be that the institutions involved will have to curtail their lending and/or acquire additional reserve assets. This will result in higher interest rates and a reduced demand for loans that, in turn, will curb the rate of growth of the money supply.

By changing the fraction of deposits that banks are obliged to keep as reserves, the central bank can control the money supply. This fraction is generally expressed in percentage terms and thus is called the *required reserve ratio*: the higher the required reserve ratio, the lower the amount of funds available to the banks. Vice versa, the lower the reserve ratio required by the monetary authorities, the higher the amount of funds available to the banks for alternative investments (see Figure 6.5).

The advantage of reserve requirements as a monetary policy tool is that they affect all banks equally and can have a strong influence on the money supply. However, the latter can also be a disadvantage, as it is difficult for the authorities to make small changes in money supply using this tool. Another drawback is that a call for greater reserves can cause liquidity problems for banks that do not have excess reserves.⁴ If the authorities regularly make decisions about changing reserve requirements it can cause problems for the liquidity management of banks. In general, an increase in reserve requirements affects banks' ability to make loans and reduces potential bank profits because the central bank pays no interest on reserves.

In some countries, such as in the United States, Japan and eurozone, central banks use reserve requirements now and again as a monetary policy tool. However, in reality they are rarely used compared with OMOs and the discount window. In fact, various central banks (such as those in Canada, Mexico, Sweden and the UK) have eliminated them entirely.⁵ The main reason for this is that the application of reserve ratios discriminates against banks (as other financial firms do not have to hold reserves at the central bank). Also, many banks may have internal resources well in excess of minimum reserve requirements and so any call to

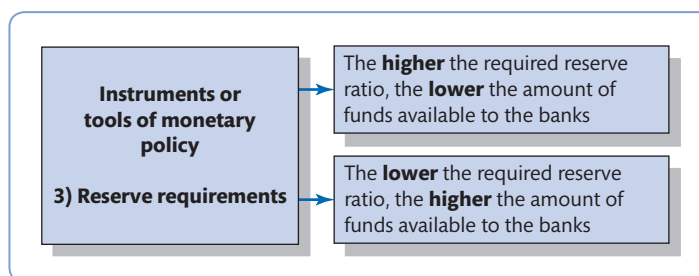


Figure 6.5 Reserve requirements

⁴ 'Excess reserves' can be defined as additional reserves that banks decide to hold over the present mandatory requirements.

⁵ (Data correct as of September 2019).

increase reserves from the authorities can easily be achieved without affecting bank behaviour in a significant fashion.

In the United Kingdom, compulsory reserve requirements were used during most of the post-war period but were abandoned as a monetary policy tool in the late 1970s especially as a result of the processes of deregulation and innovation. There are, however, compulsory *cash ratio deposits (CRD)* – not used for monetary policy purposes – determined by the Bank of England that are non-interest bearing deposits of eligible institutions (i.e. banks and building societies) that have reported average liabilities greater than £600m over a calculation period. The level of CRD is calculated twice-yearly (currently in May and November) at 0.30 per cent of average eligible liabilities (see Box 6.4).

Reserve requirements are often referred to as **instruments of portfolio constraint**. It means that they may be imposed by the authorities on the portfolio structure of financial institutions, with the purpose of influencing credit creation and, possibly, the type of lending taking place. Other instruments of portfolio constraint that are potentially available for use include: special deposits, moral suasion and direct controls.

6.4.4 Other instruments of portfolio constraint

6.4.4.1 Special deposits

In the United Kingdom, special deposits are deposits that the Bank of England may require from certain banking institutions. These deposits, equal to a specified proportion of certain elements of bank's deposit liabilities, are then 'frozen' at the Bank of England and may not be used as part of the reserve asset base for lending purposes. While they are particularly discriminatory as regards the institutions to which they apply, they do have a very rapid impact upon the ability of these institutions to create credit and are useful for drawing off any excess reserve assets within the system. At present, they are not used for monetary policy purposes. During the 1970s the Bank of England used a scheme known as 'supplementary special deposits'. These were deposits that banking institutions had to make at the Bank of England if the growth rate of some interest-bearing deposit liabilities exceeded an upper limit set by the Bank. The Bank of England operated this mechanism, referred to as the 'corset', on a periodic basis between 1973 and 1980. Institutions exceeding the specified growth rate for liabilities were required to make supplementary special deposits with the Bank on a scale dependent upon the extent of the overshoot. Supplementary special deposits could not be used as part of the reserve asset base by banks, and they attracted no interest payment. The main objective of special deposits is to remove excess liquidity from the system if bank deposit growth (and therefore loan growth) is increasing too rapidly.

6.4.4.2 Moral suasion

Moral suasion refers to the range of informal requests and pressure that the authorities may exert over banking institutions. The extent to which this is a real power of the authorities relative to direct controls is open to question, since much of the pressure that the authorities would exert involves the institutions having to take actions that might not be in the banks' commercial interests. However, the position and potential power of the authorities probably provides them with some scope to use moral suasion, which may perhaps be utilised most effectively in the context of establishing lending priorities rather than absolute limits to credit creation.

6.4.4.3 Direct controls

Direct controls involve the authorities issuing directives in order to attain particular intermediate targets. For example, the monetary authorities might impose controls on interest rates payable on deposits, may limit the volume of credit creation or direct banks to prioritise lending according to various types of customer. Although these direct controls have the benefits of speed of implementation and precision, they are discriminating towards the institutions involved and are likely to lead to disintermediation as both potential borrowers and potential lenders seek to pursue their own commercial interests. Their use, therefore, is perhaps best reserved for short-term requirements not least since their effectiveness will tend to decline the longer they are applied. Such controls, however, are widely used in many developing countries where the authorities may force banks to (say) lend a certain percentage of loan-book to 'priority sectors'.

6.4.4.4 The decline in use of portfolio constraints

In contrast to market intervention instruments (OMOs and standing facilities), instruments of portfolio constraint tend to have a narrow and therefore distorting impact. Although in principle they could be applied to a wide range of financial institutions, in the past in the UK and in the United States they have been applied only to banking institutions. The result of this is that banks have, in effect, been discriminated against. It should also be noted that disintermediation has often occurred as potential borrowers have sought alternative sources of funds, from outside of the monetary control regime.

As we noted in the case of reserve requirements, there has generally been a decline in the use of portfolio constraints as a tool of monetary policy in the world's largest economies (although they are still prevalent in many developing countries).

Portfolio constraints are less widely used in the developed world for the following reasons:

- Deregulation and increasing competition in the provision of financial services and products traditionally offered by banks have broadened considerably the number and type of institutions that would need to be brought within the control regime. Defining and implementing effective portfolio constraints would be difficult and open to controversy.
- Disintermediation, primarily involving large companies, has undermined portfolio constraints. By contrast, the use of market intervention allows the authorities to influence all relevant parts of the financial system, with monetary control coming via the price of credit.
- Throughout the 1950s, 1960s and 1970s, when portfolio constraints were used extensively, most countries maintained a system of foreign currency exchange control. This restriction on the movement of funds to and from abroad prevented borrowers from seeking finance from overseas when domestic monetary policy was restrictive. These exchange controls had the effect of supporting the portfolio constraints but (for many countries) they now no longer exist, with the result that domestic borrowers have much greater scope to seek funds from overseas.
- Portfolio constraints are regarded as inimical to competition because they place restrictions, of one kind or another, on the business freedom and growth of banks and other intermediaries falling within the constraints. Also, markets are distorted and economic efficiency tends to be undermined.

6.4.4.5 Monetary policy: transmission mechanism, fixed rules and discretion

How does the monetary transmission mechanism work?

The **monetary transmission mechanism** consists of the several channels through which policy-induced changes in money supply and interest rates influence real economic activity and in particular the price level. Understanding the channels through which monetary policy actions and strategies impact output and inflation and the time needed for a policy action to be effective is crucial for the design and implementation of sound monetary policy strategies (Figure 6.6).

The transmission mechanism of monetary policy includes several different channels. Figure 6.7 shows that changes in the official interest rates first alter money market rates and the public's expectations. These in turn affect bank rates, asset prices, the availability of bank loans and the foreign exchange rate.

Specifically:

- 1 The first indirect effect of a change in official bank rate is on the cost of borrowing. Changes in retail banks' rates for conventional financial products like mortgages, loans and deposits rates have an impact on consumption and investment.
- 2 Another channel is through prices of financial and other assets. This is also known as wealth channel because higher interest rates depress bond, stock and real estate prices, thereby decreasing households' wealth.
- 3 A third channel is money and credit availability as higher interest rates increase borrowers' default risk so households and businesses may be subject to credit rationing, thus reducing their consumption and investments.
- 4 Finally, in an open market changes in the official policy rate affect foreign exchange markets. For example, in the UK, an increase in interest rates relative to rates in other countries makes the pound stronger and more attractive to investors while British products become more expensive so their demand drops.

Figure 6.7 also shows that eventually changes in policy rate influence the overall spending in the economy i.e. the aggregate demand (AD). This is the sum of consumers' spending (C), firms investments (I), government spending (G) and net exports (NX) so that $AD = C + I + G + NX$. The general price levels are affected by the relationship between AD and the ability of the economy to supply them (aggregate supply, or AS). If $AD > AS$

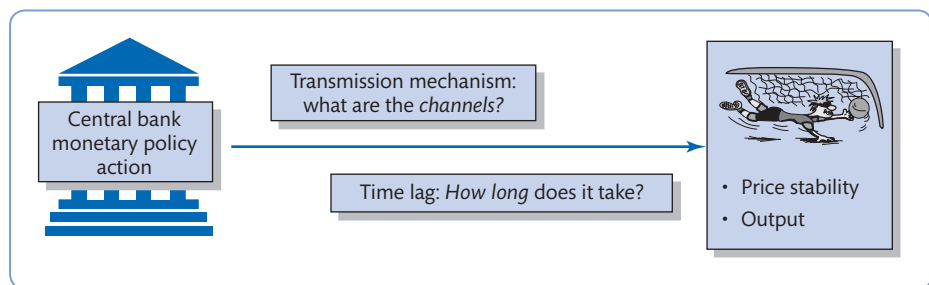


Figure 6.6 The key issues in monetary policy transmission

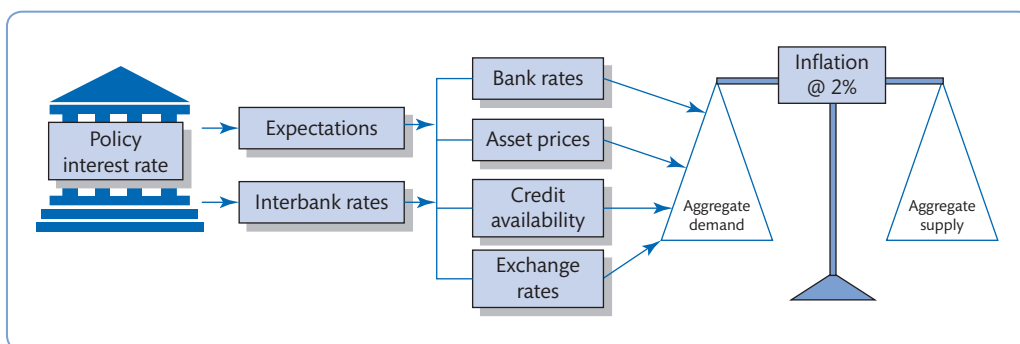


Figure 6.7 Channels of monetary policy transmission

then prices will increase so the monetary authorities may intervene to raise the policy rate to reduce growth and avoid that inflation goes above the government target (currently 2 per cent in the UK). Vice versa, if $AD < AS$ monetary authorities may be inclined to ease monetary policy to stimulate spending and keep inflation from falling below the target. To sum up changes in monetary policy ultimately alter the propensity and the ability to consume and invest of households, businesses, the government and foreign economic agents and shift aggregate demand and supply. *How long* it takes for monetary policy actions to impact output and price levels depends on a number of factors, including (1) public expectations on long-term interest rates and inflation and (2) the extent to which markets have been able to anticipate monetary policy actions and factor in future price increases.

While changes in policy rates affect relatively quickly bank rates, asset prices and exchange rates, the impact of monetary policy actions on output can take anything from three months to two years; and on inflation from one to three years or more (Figure 6.8). This is why the central banks' decisions typically have to be made with a view to the future, taking into account associated uncertainty.

How is monetary policy implemented?

Monetary authorities operate with almost complete discretion in the conduct of monetary policy. This means that although they are expected to achieve various goals in terms of sustainable economic growth, employment and price stability, current institutional settings enable them to freely weigh and pursue these goals using their own judgement.

For example, the committee in charge of setting the US monetary policy (the Federal Open Market Committee, or FOMC) meets regularly eight times a year to evaluate the current state

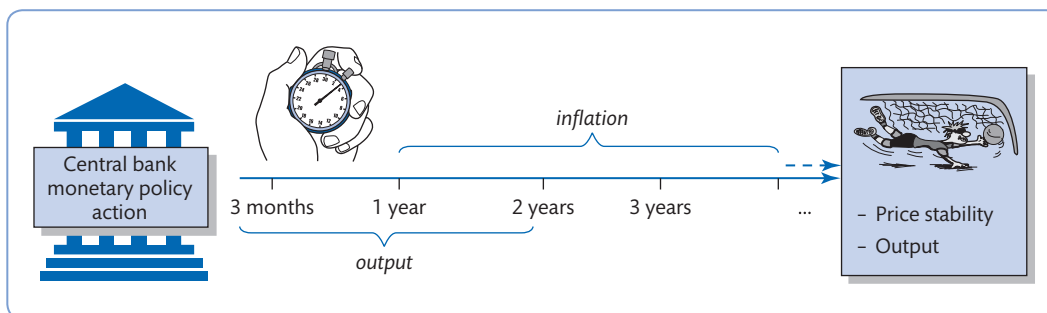


Figure 6.8 Transmission lags of monetary policy

of the economy and assess the forecast of future economic and financial conditions. Based on these considerations, the FOMC sets the Federal fund rate and then the Fed adjusts the money supply to reach that interest rate target.

Some economists advocate that monetary policy made by a precise, legislated *rule* would be more successful in ensuring a stable macroeconomic environment than using discretionary powers. A classical example of a tight, fixed rule is Friedman's constant money ($k\%$) growth rule.⁶ As pointed out by Bernanke and Reinhart (2004), although they have never been implemented, 'rule-like' monetary policy arrangements have existed in the real world, such as the international gold-standard.

Although it is not in the remit of this textbook to go into the details of the rules vs. discretion debate over monetary policy, it is useful to summarise some of their most important pros and cons (Table 6.4).

The time inconsistency problem viz. 'eat your soup or no TV'

One of the most important problems of discretion is the time inconsistency that was analysed first by economics Nobel prize winners Finn Kydland and Edward Prescott in the article 'Rules rather than discretion: The inconsistency of optimal plans', published in the *Journal of Political Economy* in 1977. Time inconsistency of policy occurs when monetary authorities decide to announce *ex ante* the policy they want to pursue to influence the expectations of private economic agents. However, *ex post*, i.e. when these latter have acted based on their expectations, monetary authorities may be tempted to revoke their decision. The outcome of this situation is that when private economic agents realise that monetary authorities' may be inconsistent over time, they lose their trust in policy makers' announcements. In order to gain credibility for their announcements, policy makers may find it optimal to commit to a fixed policy rule.

Examples of time inconsistencies can be found in different scenarios, even in relation to parenting, where the parent (policy maker) announces a punishment for bad behaviour for a child (the economic agent), e.g. 'eat your soup or no TV'. The purpose of the threat is to

Table 6.4 Pros and cons of discretion vs. fixed rules in monetary policy

	Discretion	Fixed rule-based policy
Pros	- Flexible and adaptable to changing economic conditions - Active approach that works without rules - Policy makers can respond quickly to events	- Transparent and predictable - Removes the active role of the policy maker - Leave less room for policy makers' mistakes
Cons	- Time inconsistency 'trap' and trend to forbearance especially during downturn - Long lags between problem observation and policy implementation - Unclear what monetary authority should do in case of unexpected shock - Works only if the public trust the policy maker - Requires accurate knowledge and measure of current and future economic conditions (forecast)	- Too mechanical and impractical - Passive approach that lacks flexibility in case of unexpected shock - If too strict it cannot accommodate special circumstances and it may become inapplicable - Requires a high degree of confidence that the variables will perform as expected - Need 'escape clause' in case of exceptional events (such as a war) and thus could be liable to abuse

⁶Friedman (1968). There are other well-known rules such as the Taylor rule and the McCallum rule. For more details Bernanke and Reinhart (2004) and Mishkin (2011).

affect the expectations of the child and therefore their behaviour. Although there is a clear threat, the parent may have incentives to give concessions to the child. However, the parent will be better off sticking to the behavioural rule of 'no TV' as the best outcome for the parent is to never relent to avoid undermining their credibility and prevent time inconsistency. The corollary is that policy makers can better achieve their goals by having the discretion taken away from them. In the monetary policy practice, one way of approximating this outcome might be to give the policy maker (central bank) a simple rule which is a reasonable approximation to the optimal rule. In this way if the central bank wants to deviate from this rule by more than some pre-specified amount, it is allowed to do so, but at the cost that it must explain its rationale (e.g. Athey *et al.*, 2005). Referring back to Table 6.4, it is clear that there are a number of pros and cons of the two extreme cases of pure discretion and fixed rules. This is why many countries have adopted hybrid regimes of 'constrained discretion' in monetary policy that combine elements of both fixed rules and discretion.⁷

A classic example is inflation targeting that involves 'a strong, credible commitment by the central bank to stabilise inflation in the long run, often at an explicit numerical level, but also allows for the central bank to pursue policies to stabilise output around its natural rate level in the short run.' (Mishkin, 2011). The key requirements for sound inflation targeting policies are that central banks (1) have some degree of independence and (2) are committed (and allowed) not to target other indicators (e.g. employment or exchange rates). In the case of inflation targeting, the 'constrained discretion' framework combines two distinct elements: a precise numerical target for inflation in the medium term and a response to economic shocks in the short term. For more information on inflation targeting, see e.g. Jahan (2017).

BOX 6.4 QUANTITATIVE EASING IN JAPAN

The first experiment of quantitative easing was pioneered in Japan in the first half of the 2000s following what has been defined as a 'lost decade' of economic stagnation. Japan's problems started in the 1980s when the economy experienced an unsustainable boom – a 'bubble economy' – accompanied by several 'mistakes' in the monetary policy decisions. (For a thorough discussion of the Bank of Japan's monetary policy options during the asset price bubble see Ito and Mishkin (2004).) During the boom period banks expanded considerably their lending activity to riskier customers in different industrial sectors, including real estate and construction, and small and medium businesses. Banks' lending decisions were typically based on collateral requirements and, unsurprisingly, a high proportion of these collateral were real estate (Kanaya and Woo, 2000).

When the bubble burst in the early 1990s, non-performing loans had reached alarming levels and Japan entered a period of virtually zero growth and high deflationary pressures. The highly overvalued stock and real estate markets fell dramatically, and bad loans, capital erosion and a slow policy response resulted first in the failure of small- to medium-size financial institutions in the first half of the 1990s, then a major banking crisis in 1997–1998, that saw the collapse of several large Japanese banks. Monetary authorities resorted to unconventional

⁷ Most countries' Central Banks provide illustrations of the transmission mechanism of monetary policy. These documents are often available from the Central Banks' websites. For example, see Bank of England ('The transmission mechanism of monetary policy', www.bankofengland.co.uk/markets/bank-of-england-market-operations-guide/our-objectives; European Central Bank ('Transmission mechanism of monetary policy', www.ecb.europa.eu/mopo/intro/transmission/html/index.en.html) and Bank of Canada ('How monetary policy works: The transmission of monetary policy', www.bankofcanada.ca/core-functions/monetary-policy/).

Box 6.4 Quantitative easing in Japan (*continued*)

monetary policies when the Bank of Japan official rate (the ‘uncollateralised call rate’) was lowered to virtually zero in 2001 and yet failed to reverse the process.

The Bank of Japan changed its main operating target for money market operations from the uncollateralised call rate to the current account (reserves) balances held by commercial banks, in excess of the required reserves. Initially the target amount of current account was set at 5 trillion yen, while the required reserves were about 4 trillion yen. By January 2004 the target amount has been raised in several steps to a range of 30–35 trillion yen and remained unchanged until the quantitative easing policy was terminated in March 2006. The Bank also expanded the amount of monthly outright purchase of long-term Japanese government bonds, from 400 billion yen to 600 billion yen in August 2001, and in several steps to 1,200 billion yen in October 2002. In addition, purchases of some of private debts, including asset-backed securities (ABS), as well as stocks held by financial institutions, have been introduced (Ito and Mishkin, 2004; Shiratsuka, 2010).

Although back in 2001 the Bank of Japan committed to maintain these policies until the core inflation stopped declining, many economists reckon that quantitative easing did not provide a sufficient stimulus to aggregate demand to overcome the risk of persistent consumer price deflation. In contrast, some analysts (Bowman *et al.*, 2011) argue that quantitative easing policies have had some (albeit small) positive effects on Japan’s economy but this was weakened by various factors, not least severe weaknesses in the banking sector.

6.4.5 Unconventional tools of monetary policy and the ‘zero-bound’

In ‘normal’ times, monetary policy aims at setting a target for the overnight interest rate in the money market and adjusting the supply of central bank money to that target through open market operations. In doing so, the central bank controls the liquidity levels in the market without having to lend to the private sector or the government. But what if interest rate and money supply, i.e. the ‘conventional’ monetary policy tools, are not effective in providing monetary stimulus to the economy and in limiting inflationary forces? What if interest rates reach zero or near zero levels and although the economy is struggling the monetary authorities cannot cut interest rates any further?

If a central bank reaches the ‘zero-bound’ i.e. cannot cut policy rates any further, then ‘unconventional’ or ‘non-standard’ policies may be used to try to spur lending and spending in the economy. In general terms this can be achieved by (1) providing assurance to financial investors that short rates will be lower in the future than they currently expect; (2) changing the relative supplies of securities (such as Treasury notes and bonds) in the marketplace by shifting the composition of the central bank’s balance sheet; and (3) increasing the size of the central bank’s balance sheet beyond the level needed to set the short-term policy rate at zero (*‘quantitative easing’*).⁸

Quantitative easing can be defined as an unconventional monetary stimulus designed to inject money directly into the economy to counterbalance a sharp fall in aggregate demand as spending has declined. In practical terms, the central bank purchases assets (mainly government bonds) from private business using new money created electronically. The institutions selling those bonds (such as commercial banks, other financial intermediaries and

⁸ See Bernanke and Reinhart (2004).

non-financial firms) will have new money in their account that is expected to stimulate the money supply.

In the United Kingdom the MPC introduced quantitative easing for the first time in history in March 2009, when the bank rate was reduced to 0.5 per cent. The series of asset purchases undertaken between March 2009 and February 2012 injected money directly into the system with the aim of boosting nominal demand and giving a monetary stimulus to the economy. Similarly, in an emergency meeting in March 2020, at the peak of the Covid-19 crisis, the bank rate dropped to its all-time low of 0.1 per cent and revived QE with the announcement of £200 billion of bonds emergency purchases, bringing the total to £645 billion. Table 6.5 summarises the purchases of government debt ('gilts') as well as corporate bonds authorised by the MPC since 2009 as part of the quantitative easing programme to date (December 2012). The Bank of England purchases gilts from private investors (typically non-bank financial intermediaries) that have an interest in selling low yielding assets i.e. gilts, so that they can purchase other assets like corporate bonds and shares. This process should lower longer-term borrowing costs and encourage the issuance of new equities and bonds thus improving the functioning of corporate credit market (Joyce *et al.*, 2011). While both the Japanese and UK's quantitative easing experience has implied the use of both sides of the central bank balance sheet, the US Federal Reserve bank policy appears to focus more on the asset side, a fact that explains why it is sometimes referred to as credit (rather than quantitative) easing. The US Federal Fund rate was lowered to near zero levels in December 2008, amid the financial crisis and economic slowdown. At that time, the Federal Reserve started the first programme of large scale asset purchases, which ended in March 2009 and reached \$1.75 trillion (see Table 6.5).

In theory, a narrow definition of quantitative easing implies an increase in the size of the central bank balance sheet without changing its composition. Conversely **credit easing** (narrowly defined) corresponds to changes in the composition of the balance sheet without changing its size, i.e. by replacing conventional assets with unconventional ones. Shiratsuka (2010) argues that, in practice, to be an effective non-standard policy, both sides (assets and liabilities) of the balance sheet should interact *and* the two elements of the balance sheet (size and composition) should be combined.

Table 6.5 UK and US asset purchase programmes (in figures)

Month/year	UK	US
December 2008		\$1.75 trillion
November 2009	£200 billion	
November 2010		\$600 billion
October 2011	£75 billion	
February 2012	£50 billion	
July 2012	£50 billion	
September 2012		\$85 billion/month
August 2016	£60 billion	
October 2019		\$60 billion/month
March 2020	£210 billion	\$700 billion

Note: In September 2011 and until December 2012 the US Fed launched the so-called 'operation twist' or maturity extension programme, aimed at amplifying the effects of QE on longer-term interest rates.

Source: Central banks' websites.

Generally speaking, the ECB's actions in the aftermath of the 2007–2009 global financial crisis were similar to those implemented by the BoE and the Fed as they were primarily aimed at extending the range of available facilities and designing novel mechanisms to ensure that banks could easily access the liquidity they needed. Since August 2007 the ECB has engaged in exceptional monetary policy measures known as *enhanced credit support* to stimulate the economy of the eurozone states (ECB, 2010b). The ECB's main objectives have been to increase liquidity and to improve the functioning of money markets in the euro area. Importantly, the ECB has broadened the scope of its standard refinancing operation by using 'fixed rate full allotment' provisions (rather than variable rate tender procedures) with a longer maturity of maximum one year. With these operations, the interest rate is set in advance and the ECB provides as much liquidity as the banks require provided they offer suitable collateral.

However, with the sovereign debt crisis that developed in late 2009 (see Chapter 14 for more details of the eurozone crisis) the ECB took a different approach to the Fed's and BoE's easing programmes and implemented first a **Securities Markets Programme** (SMP) in spring 2010 and then a set of **longer-term refinancing operations** (LTROs) in December 2011. In addition, the ECB introduced two programmes in 2009 and 2011 to purchase **covered bonds** (covered bond purchase programme, or CBPP) for a total of €100bn. Covered bonds are debt obligations secured by a cover pool of high-quality collateral (mortgage loans or public sector debt) that the issuer is required to maintain on balance sheet, as opposed to asset-backed securities where the credit risk can be transferred and the assets are taken off balance sheet (ECB, 2008a). The aims of the covered bond purchase programme are (1) to ease funding conditions for banks and firms and (2) to stimulate bank lending.

In mid-2014, the ECB initiated an asset purchase programme (APP) to support the monetary transmission mechanism and provide the amount of policy accommodation needed to ensure price stability. It included the purchase programmes of the corporate sector (CSPP), public sector (PSPP), asset-backed securities (ABSPP) and the covered bonds (third round, CBPP3). In addition, due to Covid-19 the ECB also initiated on 26 March 2020 a temporary pandemic emergency purchase programme (PEPP) with an overall envelope of €750 billion. All asset categories eligible under the existing APP described above are eligible for the PEPP. The total cumulative net purchases at the end of April 2020 was €2,822 billion with most of the stock (over 80 per cent) in PSPP.

Figure 6.9 reports the exceptional growth in central banks' balance sheet over the period 1999–2019 (panel a). It also shows that interest rates fell sharply both in nominal terms and became negative in real terms, with negative-yielding government and corporate bonds peaking at around \$17 trillion or 20 per cent of world GDP (panel b). Borio (2019) argues that this is the result of central banks' strenuous efforts to push a stubbornly low inflation back towards the 2 per cent targets. Finally, panel (c) shows that both private and public debt is now significantly higher as a ratio of GDP than pre-crisis.

Other forms of unconventional monetary policy in addition to quantitative easing are **forward guidance** and negative interest rate policies. Forward guidance is when the central bank communicates its future monetary policy intentions based on its assessment of the outlook for price stability. In practice, the central bank makes a public statement about the future path of interest rates, hence influencing market expectations and reducing uncertainty. There are three types of forward guidance that have been used by central banks in recent years (Ehrmann *et al.*, 2019). The first type, open-ended forward guidance, is a purely qualitative statement about the policy path. An example is the ECB's statement used

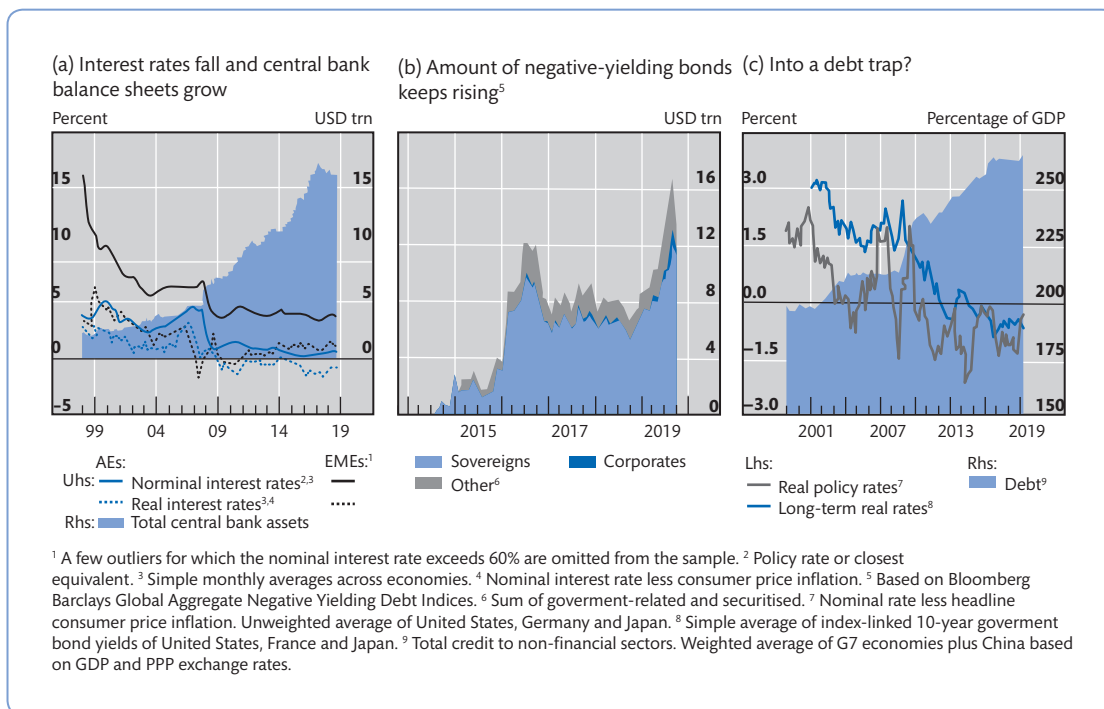


Figure 6.9 The room for monetary policy manoeuvre keeps narrowing

Source: Borio (2019).

between July 2013 and January 2016 that ‘we expect the key ECB interest rates to remain at present or lower levels for an extended period of time’. The second type, data-based forward guidance, outlines a policy path conditional on economic outcomes. An example is when the ECB stated that policy rates will remain unchanged ‘for as long as necessary to ensure the continued sustained convergence of inflation to levels that are below, but close to, 2% over the medium term’. Finally, calendar-based forward guidance is a policy path with an explicit reference to a calendar date. An example is the current guidance by the ECB that key policy rates are expected to remain at their present levels ‘at least through the first half of 2020’.

Negative interest rates have been implemented by the ECB and Bank of Japan by charging, rather than paying, interest rates on the reserves that commercial banks hold at the central bank. The expected effect is that banks will lower their excess reserves by increasing lending and purchasing other financial assets. The policy objective is to reduce lending rates, increase credit supply and boost prices across financial markets. For a detailed discussion of the non-standard monetary policy measures in practice in the eurozone, UK and Japan, see Dell’Ariccia *et al.* (2018).

6.5 Why do banks need a central bank?

The banking sectors of most countries have a pyramid structure where a central bank is at the apex and the ordinary banking institutions are at the base of the pyramid. An interesting theoretical explanation for the existence of central banks is given by Goodhart (1987). He highlights three possible reasons that might lead banks to prefer to form interbank organisations (a banking club) to which to delegate certain functions. First, the transaction and monitoring costs for interbank loans would be reduced if the central bank arranged them centrally. Second, banks would be obliged to hold a sort of ‘socially optimum’ amount of reserves that most probably would not keep otherwise. Finally, they would mitigate potential negative externalities derived by bank contagion effects.

In Goodhart’s view central banks are needed for two main reasons. First, because banks provide two essential functions: they operate the payment systems and undertake portfolio management services. These are considered to be ‘public goods’ and hence need to be preserved; meanwhile, bankers themselves have an economic interest in protecting the reputation of the banking sector as a whole and in keeping the confidence of investors. A central bank aims to prevent the collapse of the banking sector that could arise from information problems, abuses by bankers and excessive risk taking. However, Goodhart’s argument challenges the conventional view that the joint provision of payments’ services together with portfolio management functions of banks exposes the monetary system to contagious failure, which a central bank should prevent. Instead, the author maintains that it is the special nature of banks’ assets, largely non-marketable, fixed nominal-value loans of uncertain true worth that makes them more vulnerable than other non-banking firms.

Central banks can also be thought of as ‘super-banks’, at the centre of the financial system, responsible for both ‘macro’ functions, such as monetary policy decisions, and ‘micro’ functions, including the lender of last resort assistance of the banking sector. Over time the role and functions of central banks have developed and evolved, as has the environment in which banks operate. The presence of a central bank acting as a ‘super-bank’ has been investigated by the so-called ‘free-banking school’. **Free banking** theorists argue that regulation should be left to the market. Therefore, they object to a single central bank being given the ‘privilege’ – or monopoly – in issuing banknotes (Dowd, 1996).

Liberalisation, financial innovation and technology have contributed to major changes in the operating environment. Many banks have converted their status to private institutions – that by definition are demand or customer-driven – while the global riskiness and uncertainty of the sector has significantly increased. The next section focuses on the LOLR function of central banks and how this has changed following the 2007–2009 financial crisis. While there is debate about the merits and demerits of central banks it is a fact that there is a central bank in virtually every country in the world. One of the key issues is whether central banks should be given independence from governments and, if so, to what extent. These issues are covered in Section 6.6.

6.5.1 The lender of last resort function of the central bank

The lender of last resort function of a central bank is often subject to controversial debates and criticisms because it implies direct intervention of the monetary authorities in the banking markets.

In its role as a LOLR, the central bank will provide reserves to a bank (or banks) experiencing serious financial problems due to either a sudden withdrawal of funds by depositors or to a situation where the bank has embarked on highly risky operations and thus cannot find liquidity anywhere else (i.e. no other institutions will lend to a bank considered near collapse). Technically, this important function of central banks derives from the discount window tool that, as illustrated in Section 6.4.2, is one of the instruments used to influence reserves and money supply in the banking sector. However, central banks operate under different frameworks in conducting the LOLR activities. These differences can reflect various country-specific factors such as public policy objectives, historical experience or other elements.

A central bank will extend credit to an illiquid bank to prevent its failure only in exceptional circumstances and in doing so it also carries out a 'macro' function by preventing potential financial panics. However, the central bank cannot guarantee the solvency of every banking institution in a country. (On the relationship between liquidity and solvency see Box 6.5.) This is because it would encourage bankers to undertake undue risk and operate imprudently, especially if banks knew that they would always be bailed out (by taxpayers' money) were they to become insolvent. In other words, the security of the LOLR function could induce or increase moral hazard in banks' behaviour.

BOX 6.5 RELATIONSHIP BETWEEN LIQUIDITY AND SOLVENCY

As we noted earlier, the liquidity of a bank relates to its ability to meet short-term obligations (expected and unexpected) when they fall due. For example, banks can predict with a certain degree of accuracy how much cash they need to hold to meet such things as payment of utility bills (electricity, water bills), rent on buildings and (under normal market conditions) deposit withdrawals. In addition to these expected calls on liquidity, banks also have to hold a cushion above this amount to meet unexpected liquidity requirements and this is why liquidity risk management is a key feature of banking business.

Solvency is the ability of a bank ultimately to meet all its obligations. This means that the value of assets has to be greater than liabilities – the difference between the two being the bank's capital. If some assets go bad (e.g. loans are not repaid) then the bank must make charges against the loan portfolio (loan loss provisions) that are paid for from retained profits. As long as profits are sufficient to cover these provisions, the level of bank capital and its capital adequacy ratio (the Basel risk-weighted capital–asset

ratio) remain unchanged. However, when profits do not cover provisions, then losses will have to be written out of capital (the amount of capital the bank has declines) or alternatively, shareholders will be asked to provide additional capital to restore the capital ratio to the required level.

The liquidity and solvency position of a bank are related because a severe liquidity shortfall can ultimately result in a solvency problem. For example, a continuous liquidity shortage could lead a bank to tighten its lending policy, for instance, by not renewing short-term revolving lines of credit. This would force borrowers to repay their loans earlier than expected and those unable to do so will default on their loan payments. In turn, the quality of the loan portfolio will deteriorate, requiring additional loan loss provisions. If the bank does not have sufficient provisions then the losses will be written out of capital resources thus reducing the solvency of the bank. In general, if a bank is unable to meet its liquidity requirements it will first attempt to obtain support through the LOLR facility by borrowing from the central bank. However,



Box 6.5 Relationship between liquidity and solvency (*continued*)

if this option is not available then the bank will have to consider bearing the losses from its capital resources thus reducing the bank's capital position.

A liquidity crisis can quickly change into a solvency crisis. For example, if a bank has actual (or even perceived) liquidity difficulties, and this becomes known to other banks, then the latter will withdraw their wholesale interbank deposits from the illiquid bank, leading to a wholesale deposit run, which may then spread to retail deposit withdrawals. In this case the bank will not be able to meet its short-term obligations and will have to resort to using capital resources to bear any losses. Ultimately the bank may not have sufficient capital resources to meet these losses, resulting in insolvency.

In the case of an individual bank failure, the decision as to whether a bank is illiquid or insolvent is a critical judgement that the regulators have to make in order to decide whether to support a troubled bank or to let it fail. For example, if a small bank is facing financial difficulties the regulator has to decide whether this is a result of a short-term liquidity problem or a longer-term solvency issue. If the authorities

decide the problems are caused by liquidity problems, they are more likely to offer support through the lender of last resort facility (providing new liquidity to the bank to help it get through its liquidity problem). If the bank is deemed to have a solvency problem, then the authorities are more likely to consider letting the bank go bankrupt or organising some form of rescue. In theory regulators should consider supporting banks with a liquidity problem but not with a solvency problem – although we know in reality that this is not always the case. Another point to note is that regulators have a very difficult task in deciding whether a bank that is in trouble is either illiquid or insolvent as often a decision has to be made very quickly as to whether liquidity support is to be provided or not. Often regulators have to decide in a matter of days the liquidity and solvency position of a troubled bank and they will also have to take into consideration whether a decision not to provide liquidity will result in failure and serious systemic repercussions. In general, the relationship between liquidity and solvency is at the heart of bank regulation and is a critical feature of the bank supervisory process.

One of the most famous examples of the direct support given by the Bank of England to the banking sector was offered in 1973–1974 where the Bank organised help for 26 fringe (or secondary) banks facing severe liquidity problems. In the United States, famous examples of the Fed acting as a LOLR include the rescue effort of two troubled banks: Franklin National in 1974 and Continental Illinois National Bank in 1984.

Since 2007, the Bank of England, the Fed and the European Central Bank have injected substantial amounts of liquidity into their respective banking systems; these market interventions placed their actions increasingly under the spotlight. In an unprecedented move, on 1 December 2010, the Federal Reserve released previously confidential information about its function as LOLR during the 2007–2009 period, including the names of the financial institutions and foreign central banks that received financial assistance, the amounts borrowed, the dates credits were extended, the interest rates charged, information about collateral, and a description and rationale of the credit terms under each Federal Reserve emergency facility.⁹

⁹The release of data was as a consequence of Bloomberg's request for the data under the Freedom of Information Act. After a series of appeals, the Fed conceded the issue and in December 2010 it released detailed information about individual credit and other transactions conducted during the financial crisis. In accordance with the Dodd–Frank Wall Street Reform and Consumer Protection Act of 2010, transaction-level details for discount window loans and open market transactions will be made available on a quarterly basis and with an approximately two-year lag. Information is available through the Federal Reserve Board's website and the Federal Reserve Bank of New York's websites.

After Greek banks fell into arrears with the International Monetary Fund in 2015, the ECB granted the Bank of Greece a loan through the **Emergency Liquidity Assistance** (ELA). This is a last safety net provided by national central banks in the euro area for domestic banks in difficulty. The ECB's role within ELA is to check and monitor the national central banks' LOLR activities.

The depth and length of the global financial crisis transformed the role of LOLR to what is now discussed as a **market maker of last resort (MMOLR)** function. The traditional Bagehot rule of central banking (LOLR activities should be limited to solvent but illiquid banks with good collateral, to avoid moral hazard problems) has been stretched to the limit in order to meet the liquidity needs of increasingly dysfunctional financial markets. Some commentators have welcomed this transition, as it reflects the evolution of the LOLR function to meet the reality of modern financial systems but in stressed situations where there are no market buyers, the central bank acts more like a 'purchaser of last resort'. Others are more critical of the explosion of central banks' balance sheets, as well as of the collateral policies central banks followed during and after the 2007–2009 crisis. Other concerns include the inflationary consequences of this expansion of central bank credit and the lack of transparency of the decision-making process. Some have criticised central banks for bailing out insolvent institutions, and many have noted that quasi-fiscal support has been provided through generously priced bank support packages (Moe, 2012).

6.6 Should central banks be independent?

There has been a significant trend towards **central bank independence** in many countries and the issue has generated substantial debate all over the world. Theoretical studies seem to suggest that central bank independence is important because it can help produce a better monetary policy. For example, an extensive body of literature predicts that the more independent a central bank, the lower the inflation rate in an economy.

Central bank independence can be defined as independence from political influence and pressures in the conduct of its functions, in particular monetary policy. It is possible to distinguish two types of independence: goal independence, that is, the ability of the central bank to set its own goals for monetary policy (e.g. low inflation, high production levels); and instrument independence, that is, the ability of the central bank to independently set the operational instruments of monetary policy to achieve these goals (Mishkin, 2000).

It is common for a central bank to have instrument independence without goal independence; however, it is rare to find a central bank that has goal independence without having instrument independence. In the United Kingdom, for example, the Bank of England is currently granted instrument independence and practises what is known as inflation targeting. This means that it is the government that decides to target the inflation rate (at the time of writing – 2021 – set at 2 per cent) and the Bank of England is allowed to independently choose the policies that will help to achieve that goal. Such a situation is only acceptable in a democracy because the Bank of England is not elected and thus goals should only be set by an elected government.

While central bank independence indicates autonomy from political influence and pressures in the conduct of its functions (in particular monetary policy), dependence implies subordination to the government. In this latter case, there is a risk that the government may

BOX 6.6 WHO OWNS THE CENTRAL BANK?

There is no general answer to this question. It depends on the specific cases. For example, the Bank of England is 'wholly-owned by the UK government'. This means that the entire capital of the Bank is held by the Treasury solicitor on behalf of HM Treasury. Current regulations provide that the Bank is required to pay to HM 'on every fifth day of April and October, a sum equal to 25% of the Bank's post-tax profit for the previous financial year or such other sum as the Bank and the Treasury may agree'. The overall effect is that the Bank and HM Treasury will normally share post-tax profits equally (see the Bank of England Act 1946 and 1998 amendments).

While the Bank of England is owned by the British government, the ECB is owned by the central banks of member states as clearly reported in its Statute. This means that member central banks have voting powers as shareholders (article 10.3 on Governing Council) and share the profits and losses of the ECB (article 33.1). But who owns the national member state central banks then? The answer can be found in their own statutes: they are owned by the major financial institutions in a country.

Another interesting example is that of the US Federal Reserve System. Who owns the Fed? The Federal Reserve System is not 'owned' by anyone. The Federal Reserve was created in 1913 by the Federal Reserve Act to serve as the nation's central bank. The Board of Governors in Washington, DC, is an agency of the federal government and reports to and is directly accountable to the Congress. The Federal Reserve, like many other central banks, is an independent government agency but also one that is ultimately accountable to the public and the Congress.¹⁰ The 12 regional Federal Reserve Banks, which were established by the Congress as the operating arms of the nation's central banking system, are organised similarly to private corporations – possibly leading to some confusion about 'ownership'. For example, the Reserve Banks issue shares of stock to member banks. However, owning Reserve Bank stock is quite different from owning stock in a private company. The Reserve Banks are not operated for profit, and ownership of a certain amount of stock is, by law, a condition of membership in the System. The stock may not be sold, traded, or pledged as security for a loan; dividends are, by law, 6 per cent per year. This means that the concept of ownership in the case of member banks takes some unique connotations.

'manipulate' monetary policy for economic and political reasons. It should be noted, however, that all independent central banks have their governors chosen by the government; this suggests that to some extent central banks can never be entirely independent.

Since the financial crisis, the growing consensus on the positives of central bank independence has come under increasing pressure. For example, after the collapse of Lehman Brothers in 2008, the independence of the Fed has been under scrutiny. The Fed's performance during and after the 2007–2009 crisis has been severely criticised. As a consequence of growing political pressure, the Dodd–Frank Act amended Section 13(3) to require the Treasury's approval for its exercise of LOLR activities. The ECB independence has also been compromised by the adoption of the Outright Monetary Transactions (OMT) programme, which entails a promise to purchase sovereign debt of individual euro member states. The programme implies that purchases are conditional on the state meeting certain fiscal targets. While the programme has helped recover market confidence, some see it as a 'fiscalisation of monetary policy', as it allows government intervention and may lead to the loss of ECB's independence (Bullard, 2013).

¹⁰see www.federalreserve.gov/faqs/about_14986.htm

It is possible to distinguish between the case ‘for’ and the case ‘against’ independence. In general terms there are mainly political and economic reasons in support of central bank independence. Moreover, there is a principal–agent problem between the public (the principal) and the central bank and the government (the agents). The agents may have incentives to act against the interest of the public. According to the supporters of central bank independence it is the government in particular that has a strong incentive to act in its own interest. Only an independent central bank operating outside the day-to-day business of politics can be considered a guarantor of long-term economic stability. Table 6.6 summarises the arguments in favour and against central bank independence.

A survey on the future of central bank independence and inflation targeting conducted in 2017 (Den Haan *et al.*, 2017) emphasised the importance of measuring empirically central bank independence using different indicators, including the rules of appointment and dismissal of governors, the extent to which the government and tax authorities must be consulted before monetary policy decisions and the bank’s financial autonomy. Central banks are complex organisations whose mandate has expanded after the great financial crisis. As discussed in this chapter, many have adopted unconventional monetary policies, acted with a great deal of discretion and some were even given powers to go beyond their traditional monetary policy mandate in macro- and micro-prudential supervision and crisis management (see Chapter 7). In this context perhaps it is not a surprise that the debate over their independence has intensified in recent years because of the rise of populism (Goodhart and Lastra, 2018; Masciandaro and Passarelli, 2018) and many have emphasised the need for a robust governance framework and adequate mechanisms to ‘guard the guardians’ (Cochrane and Taylor, 2016; Bossu and Rossi, 2019).

Table 6.6 Central bank independence (CBI)

In favour	Against
- As monetary policy works with lags, it is necessary to take a long-term view when making decisions. CBI offers a safeguard against short-term political interests. - CBI can prevent governments from using central banks to finance government spending. Given the large public debt in many countries, CBI should act as a safeguard against deficit increases. - Political pressures can damage the credibility of a central bank, and distort inflation expectations, which goes against central banks’ ultimate goal to maintain price stability. - Many empirical studies have shown that higher levels of CBI result in economic growth and price stability. - CBI also allows central banks to move more swiftly and effectively in response to macroeconomic shocks.	- Lack of democratic governance. - Constraints of central banking (i.e. central banks may be too cautious for fear of political interference and for fear of losing independence and credibility). - Poor social outcomes (deflation).

Source: Adapted from Toyoda (2013).

6.7 Conclusion

A central bank's main function is to undertake monetary control operations and thus to administer the amount of money in the economy, given the specific monetary policy objectives set by the government. Central banks tend to use indirect instruments, or tools, in the conduct of their monetary policy and can generally choose between (1) open market operations, (2) the discount window and (3) reserve requirements. OMOs are popular instruments used by central banks as they are tax-free (in that they do not place specific constraints on banks' operations) and flexible (in terms of their frequency of use and scale of activity). In particular, the main attractions for using OMOs to influence short-term interest rates are as follows: they are initiated by the monetary authorities who have complete control over the volume of transactions; they are flexible and precise – they can be used for major or minor changes in the amount of liquidity in the system; they can easily be reversed; and finally, they can be undertaken quickly.

The financial crisis of 2007–2009 brought to the fore the key role of central banks in responding to crises. In particular, unconventional policies such as quantitative and credit easing have been key monetary policy tools in the aftermath of the global financial crisis. While the role and core functions of central banks around the world keep evolving to adapt to the transformations in the operating environment, recent debates have revolved around their greater powers and discretion particularly in emergency times. So, if there has been a trend in both developed and developing countries towards increased independence of central banks from political pressure, central bank independence has come under renewed scrutiny post financial crisis. Concerns have emerged in particular with respect to the balance between their accountability, transparency and independence, as well as the increased risk of politicisation. Hence the growth in the interest in central bank governance frameworks and in discussions over whether and to what extent central banks should abide by rule-based policy in normal times and then take a 'what-it-takes' approach with emergency powers in case of crises.

Key terms

Accountability	Forward guidance	Monetary aggregates	Price stability
Central bank independence	Free banking	Monetary policy	Prices and incomes policy
Covered bonds	Instruments of portfolio constraint	Monetary transmission mechanism	Quantitative easing (QE)
Credit easing	Lender of last resort (LOLR)	Money	Reserve requirement
Discount window	Longer-term refinancing operations (LTRO)	National debt management policy	Securities Markets Programme
Emergency liquidity assistance (ELA)	Market maker of last resort (MMOLR)	Official bank rate	Standing facilities
Exchange rate policy		Open market operations	
Financial stability			
Fiscal policy			

Key reading

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REVISION QUESTIONS AND PROBLEMS

- 6.1 What are the five major forms of economic policy?
- 6.2 What is money and what are the monetary aggregates?
- 6.3 Outline the differences between monetary policy tools, instruments and goals.
- 6.4 Why are OMOs the most popular monetary policy tool?
- 6.5 What is the Bank of England official rate?
- 6.6 Explain the meaning and limitations of the 'instruments of portfolio constraint'.
- 6.7 Discuss the key issues in the transmission of monetary policy. Briefly explain the channels through which monetary policy actions impact the economy.
- 6.8 Outline the advantages and disadvantages of discretion versus fixed-rules in monetary policy. Focus on the time inconsistency problem.
- 6.9 Define the unconventional monetary policy tools. What is quantitative easing?
- 6.10 What are forward guidance and negative interest rate policies?
- 6.11 Explain the lender of last resort function. Why is it controversial?
- 6.12 What are the arguments for and against an independent central bank?